20XX

Study customer churn analytics for the banking industry : Identify Top Three Focus Areas

Prepared By: Anubhav

Course Code: UGC | BA&C | March'25 | ENG-TXCHO4H0-20250331

Organization: UpGrad

Choosing the Right Data Points

The analysis commenced with a comprehensive banking dataset containing 10,000 customer records across three geographic regions. The dataset encompassed demographic variables including Geography, Gender, and Age, financial metrics such as Balance, Estimated Salary, and Credit Score, and behavioral indicators including Tenure, Number of Products, Credit Card ownership, Activity Status, and Exit Status.

Data point selection followed a systematic, business-driven approach:

- Prioritized variables with strong churn separation and applied bucketed analysis to those features where it added the most insight.
- Focused on metrics directly linked to revenue impact and retention outcomes.
- Leveraged SQL extensively to create %ile buckets, compute summary statistics, and perform cross-tabulation analysis to identify meaningful patterns.

Exploratory Discovery and Pattern Recognition

Initial exploratory data analysis revealed an overall churn rate of 20.37 %, representing 2,037 customers out of 10,000. However, stratified analysis exposed significant variation across customer segments:

- Germany exhibited the highest churn concentration at 32.44 %, more than double the rates observed in France and Spain at approximately 16 %
- The 56 to 65 age group demonstrated extreme churn risk at 48.32 %, representing the highest-risk demographic segment
- Inactive customers churned at 26.85 % compared to active customers at 14.27 %
- Customers holding 3 to 4 products showed anomalous churn patterns ranging from 82.71 to 100 %

Composite Score Development

Single-variable analysis proved insufficient for predictive accuracy. Three composite scoring metrics were developed to synthesize weak individual signals into powerful predictors:

Engagement Score (ES)

- Combined activity status, product holdings, and credit card ownership
- Low engagement scores correlated directly with elevated churn rates
- Provided granular insight into customer interaction patterns

Risk Score (RS)

- Integrated Germany geography, age groups 36 to 65, inactive status, and multi-product holdings
- High-risk scores identified customers combining multiple churn factors
- Enabled disproportionate churn concentration analysis

Composite Score Development

Financial Value Score (FVS)

- Synthesized account balance, estimated salary, and product depth
- Identified high-value customers at elevated risk of departure
- Enabled prioritization of retention resources based on revenue impact

Advanced Analytics Through Window Functions

Structured Query Language window functions enabled precise customer segmentation analysis. Five targeted queries using ROW_NUMBER ranking functions produced actionable customer segments:

- Top 3 highest-balance churned customers all maintained inactive status despite multiple product holdings.
- Top 5 longest-tenure retained customers demonstrated stable, active account status with consistent balances.
- Top 10 credit card holders with minimal product engagement represented significant cross-sell opportunity.
- Top 5 German customers with high balances and inactive status identified 1 million rupee revenue at immediate risk and to be offered reduced interest rates.

Three Focused Areas

Prioritization was determined by analysis of impact magnitude combined with implementation feasibility, distilled from more than 20 identified insights.

Area 1: Germany-Focused Retention Initiative

- 814 customers churned from Germany versus 810 and 413 combined from France and Spain respectively.
- Top 5 at-risk German customers maintained an average balance of 207,389 rupees, representing 1 million rupee cumulative revenue exposure.
- Implement localized retention campaigns with German-language support and priority relationship manager engagement.

Three Focused Areas

Area 2: Inactive Account Reactivation Program

- Churn differential between inactive and active customers is 12.6 %, representing 1,262 preventable departures.
- All top-5 high-balance German target customers demonstrated inactive account status.
- Deploy automated reactivation triggers and digital engagement initiatives upon activity status transition.

Three Focused Areas

Area 3: Two-Product Bundle Strategy

- Two-product customers maintained 7.58 % churn compared to 27.71 % for single-product customers and 82 to 100 % for multi-product customers.
- Four of five top German target customers already held two-product relationships.
- Execute structured cross-sell campaigns targeting single-product customers toward two-product combinations.

Three Focused Areas

Area 3: Two-Product Bundle Strategy

- Two-product customers maintained 7.58 % churn compared to 27.71 % for single-product customers and 82 to 100 % for multi-product customers.
- Four of five top German target customers already held two-product relationships.
- Execute structured cross-sell campaigns targeting single-product customers toward two-product combinations.

Business Impact and Outcomes

- By segmenting churn risk and customer value, the analysis identified Germany, inactive accounts, and multi-product complexity as priority areas for intervention.
- Targeted retention campaigns in the German segment can directly impact churn reduction, protecting up to ₹1 million in high-balance accounts and stabilizing market share in a high-risk region.
- Inactive account reactivation initiatives, such as digital engagement and automated activity triggers, have the potential to prevent approximately 1,262 customer exits, which translates to significant retention cost savings and increased lifetime value.
- Streamlined cross-sell strategies, promoting the two-product sweet spot, expand wallet share while simultaneously decreasing churn by 20 percentage points among single-product customers.
- Together, these focused actions create sustainable improvements in customer retention rates, deliver measurable revenue protection, and enhance customer lifetime value across critical banking segments.

THANK YOU!